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INT 7031 - Web Mining
Lecture 2: Web Mining Overview

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Hanoi, 11/2024

Outline

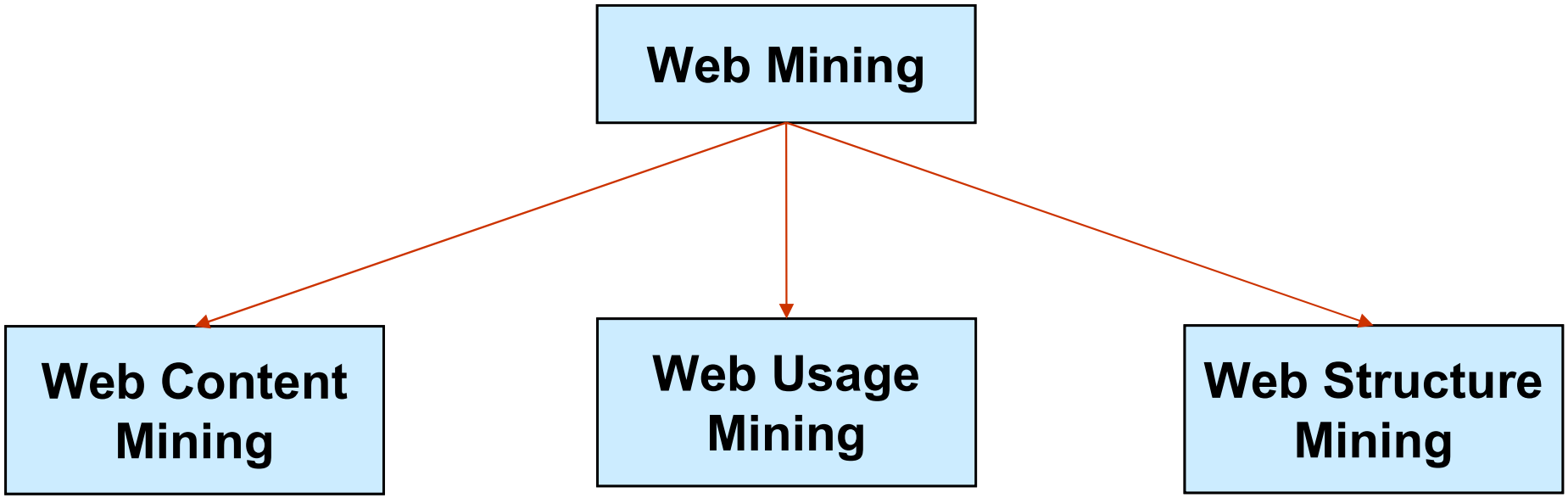
- Web Mining Overview
- Web Content Mining
- Web Structure Mining
- Web Usage Mining

What is Web Mining?

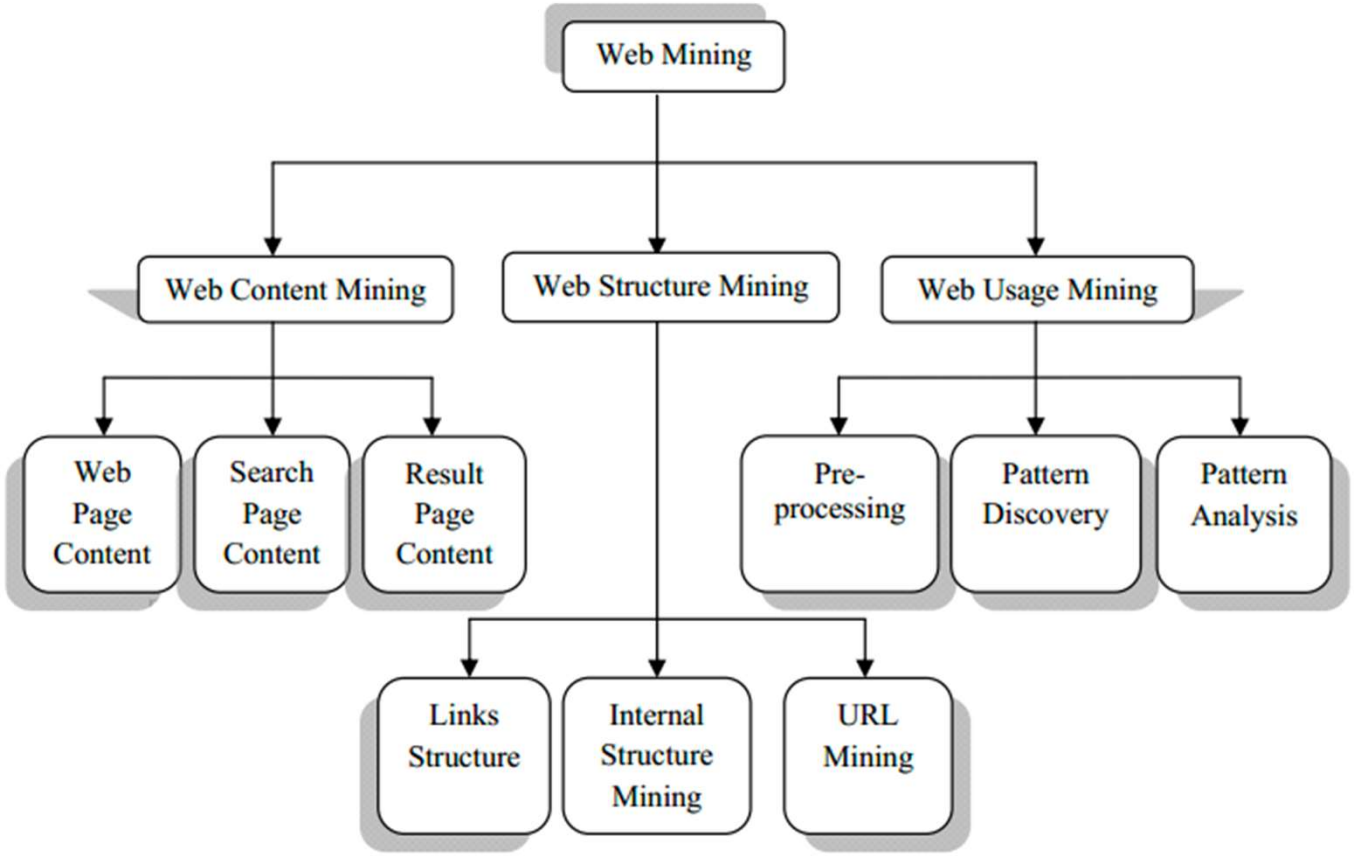
Application of ***data mining*** and ***machine learning*** techniques to extract useful knowledge from **the content, structure, and usage of Web resources.**

- Web data includes :
  - Web documents
  - Hyperlinks between documents
  - Usage logs of websites
- The WWW is huge, widely distributed, global information service centre and, therefore, constitutes a rich source for data mining.

Types of Web Mining

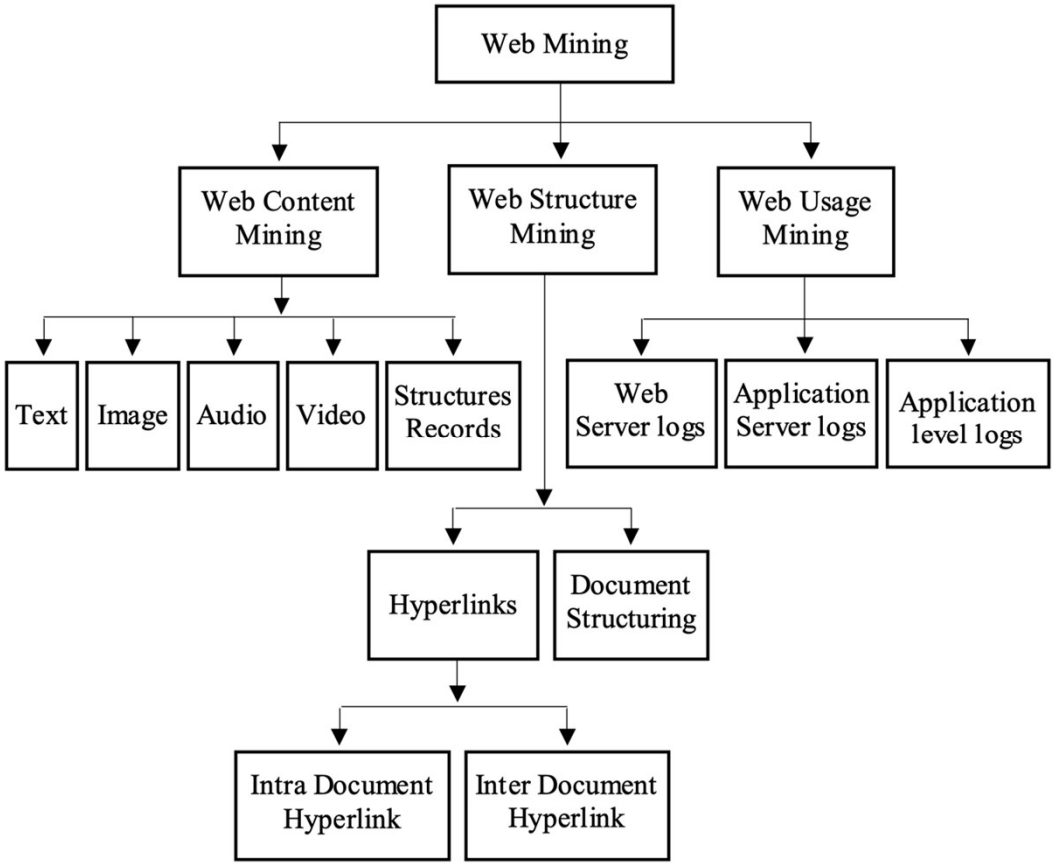


Taxonomy of Web Mining



Source: https://www.researchgate.net/figure/Taxonomy-of-Web-Mining-Source-Kavita-et-al-2011_fig1_282357293

Web Data

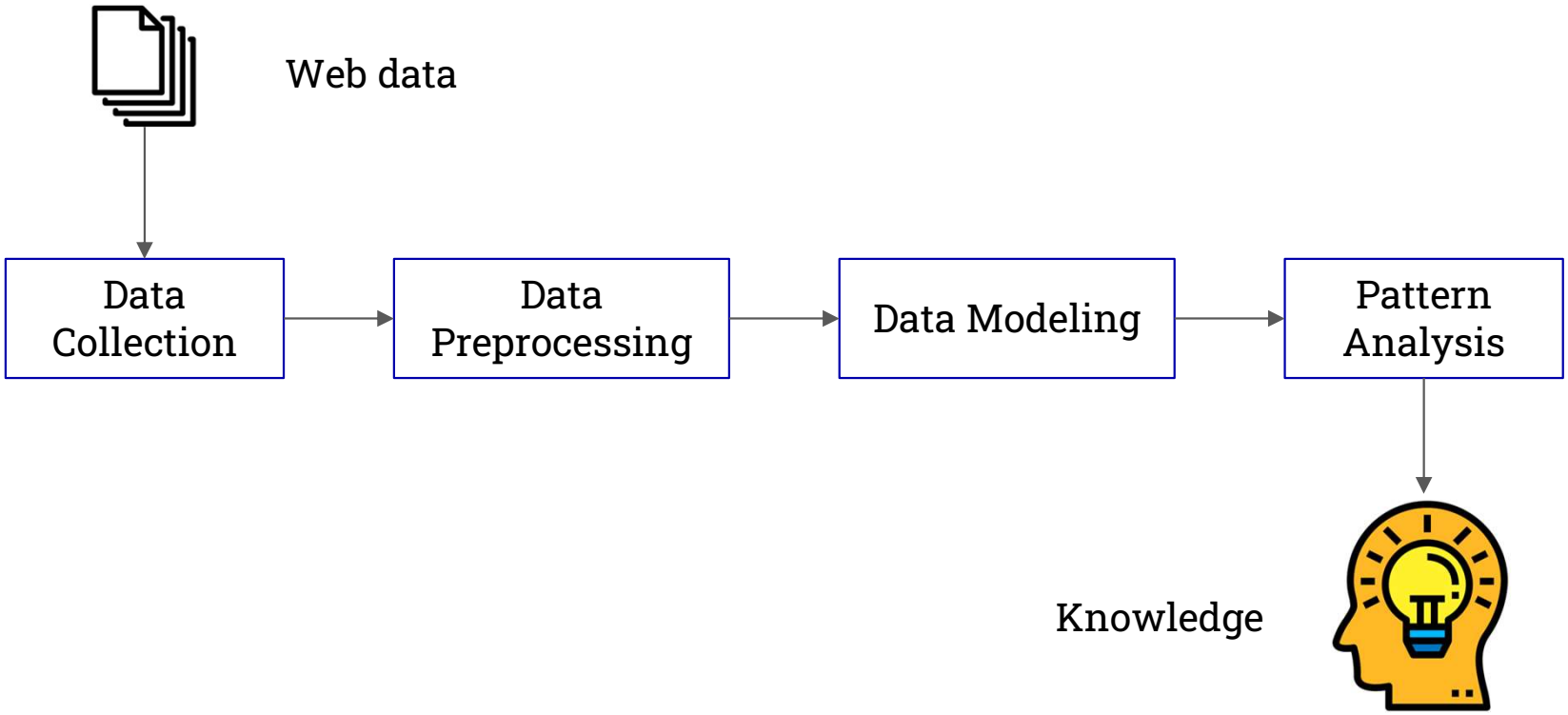


Source: Patel, Ketul B., Jignesh A. Chauhan and Jigar D. Patel. "Web Mining in E-Commerce : Pattern Discovery , Issues and Applications." (2011).

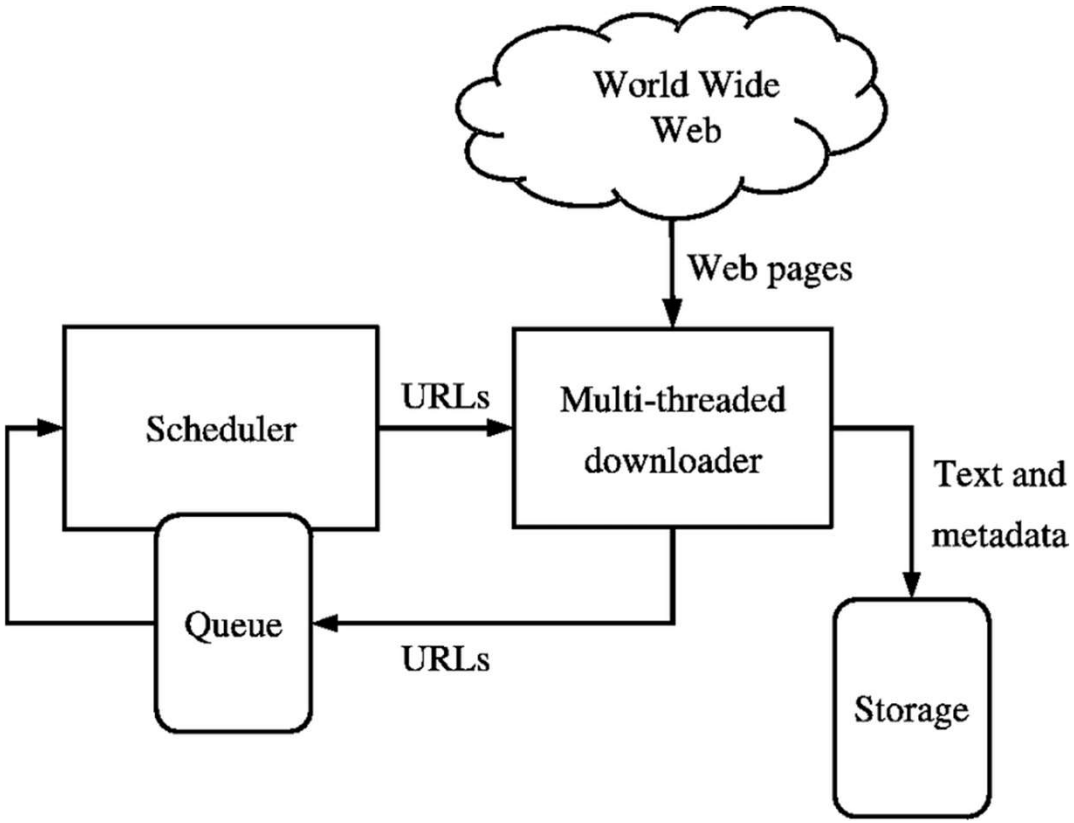
Challenges of Web Mining

- **Web is noisy :** A Web page typically contains a mixture of many kinds of information, e.g., main contents, advertisements, navigation panels, copyright notices, etc.
- **Web is dynamic :** Information on the Web changes constantly. Keeping up with the changes and monitoring the changes are important issues.
- **Web is a virtual society :** It is not only about data, information and services, but also about interactions among people, organizations and automatic systems, i.e., communities.
- Many other such restrictions pose a pretty big challenge for mining the web, e.g., big data

Web Mining Pipeline



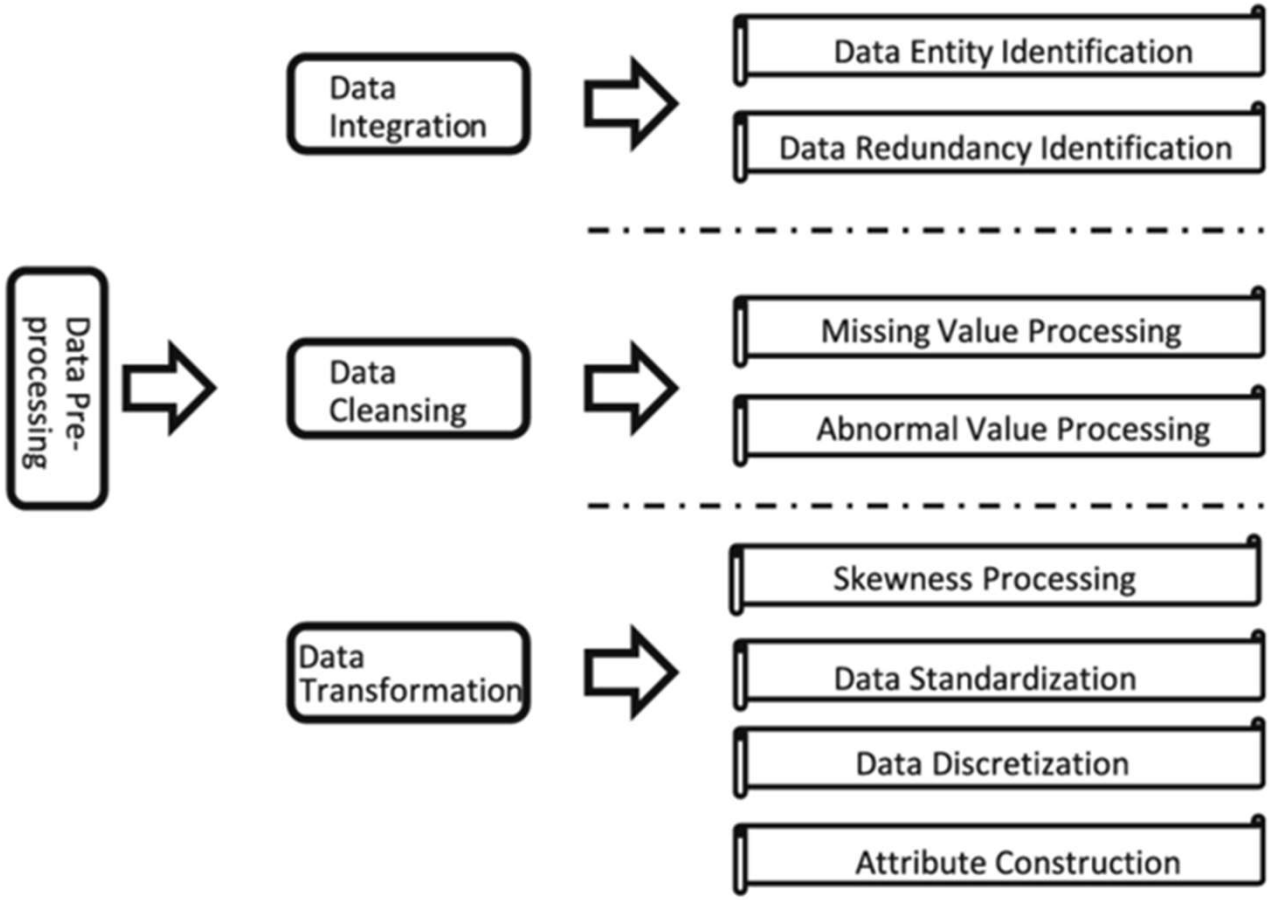
Data Collection



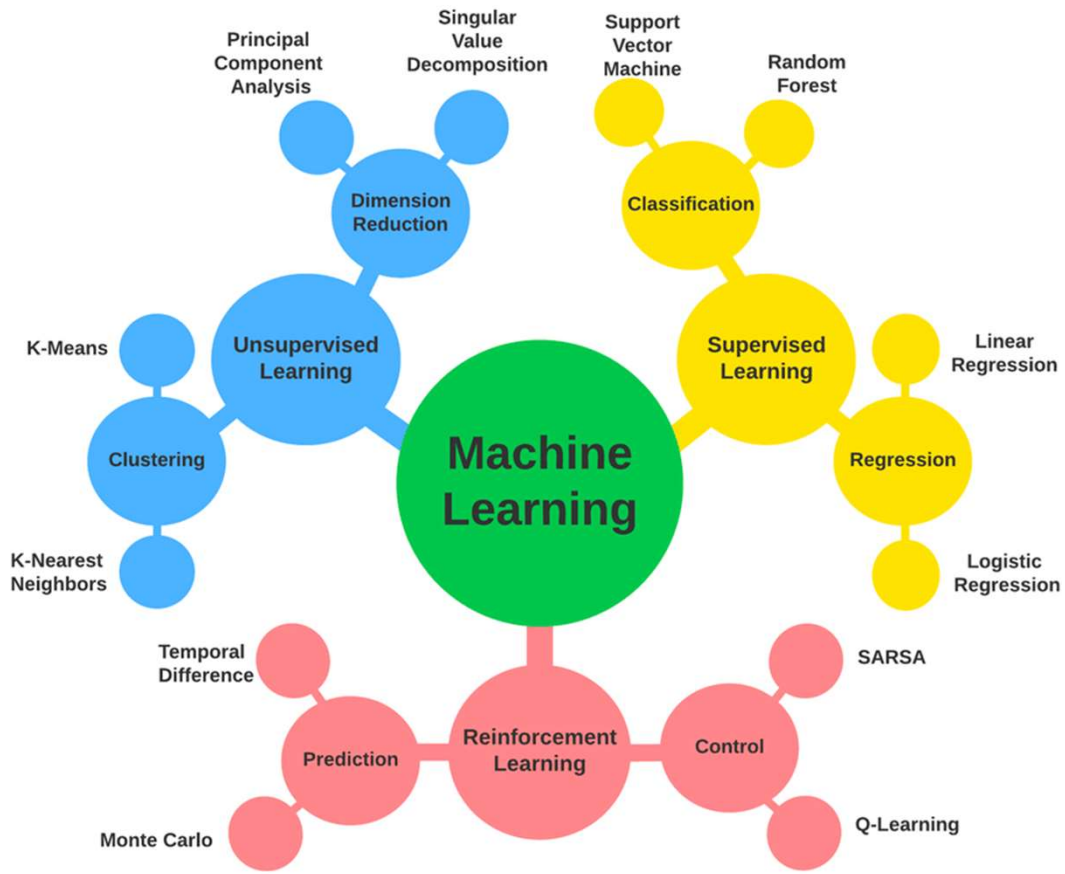
A typical data collection workflow

Data Preprocessing

A typical data preprocessing workflow

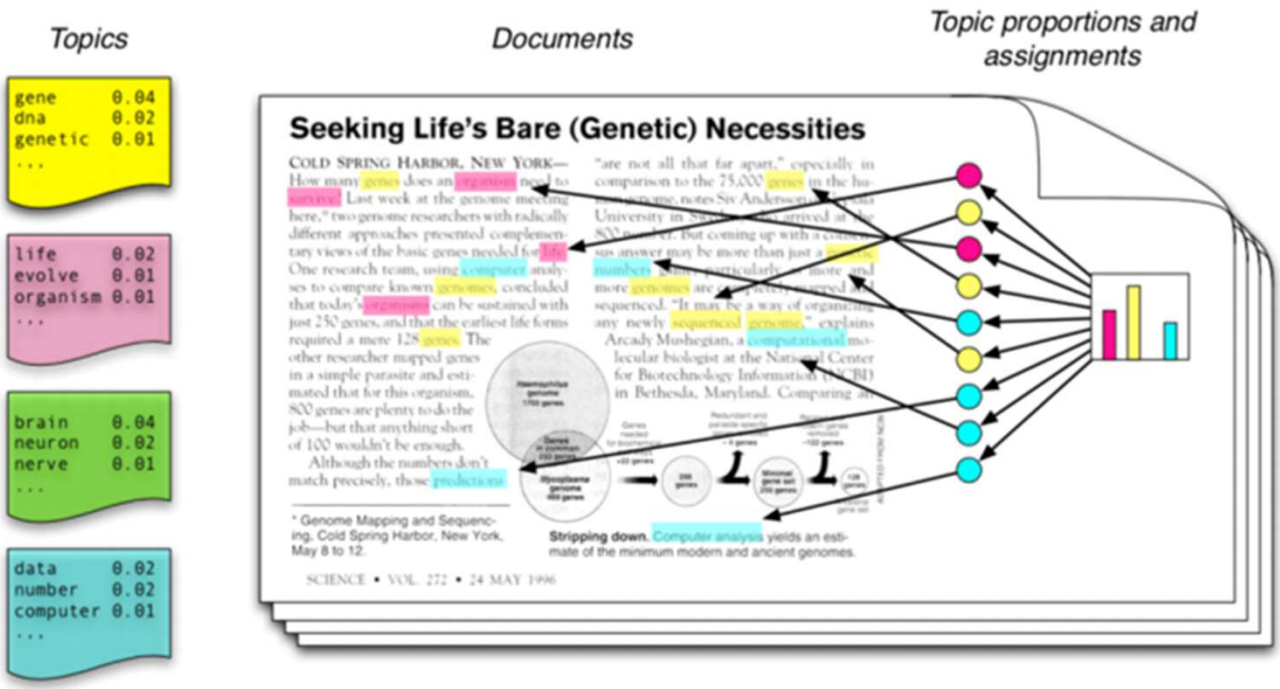


Data Modeling



A typical machine learning techniques

Pattern Analysis



Web Content Mining

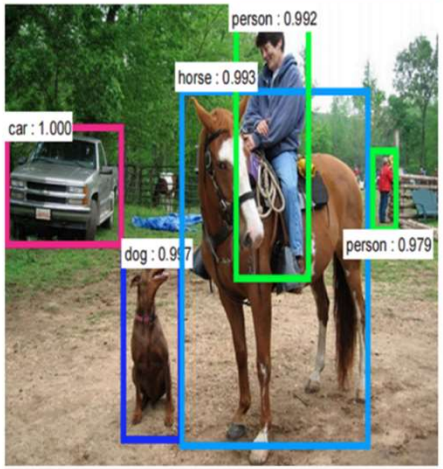
- **Goal**: Mining, extraction and integration of useful data, information and knowledge from Web page content.
- Web content mining is **related** but **different** from **data mining** and **text mining**.
- Web data are mainly **semi-structured** and/or **unstructured**, while data mining deals primarily with structured data.

```html
<!doctype html>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <title>Hello, world!</title>
    </head>
    <body>
        <h1>Hello, world!</h1>
        <p>My first web page.</p>
    </body>
</html>
```

Web Content Mining

Text

Image



Audio

Video

Web Data Mining - Text

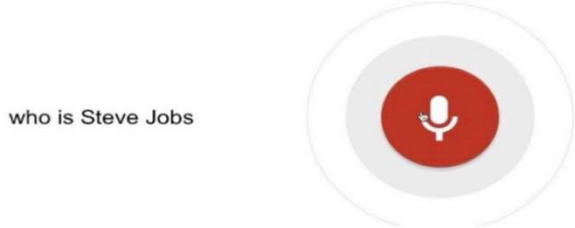
Text mining with Web Data

Spam Filtering

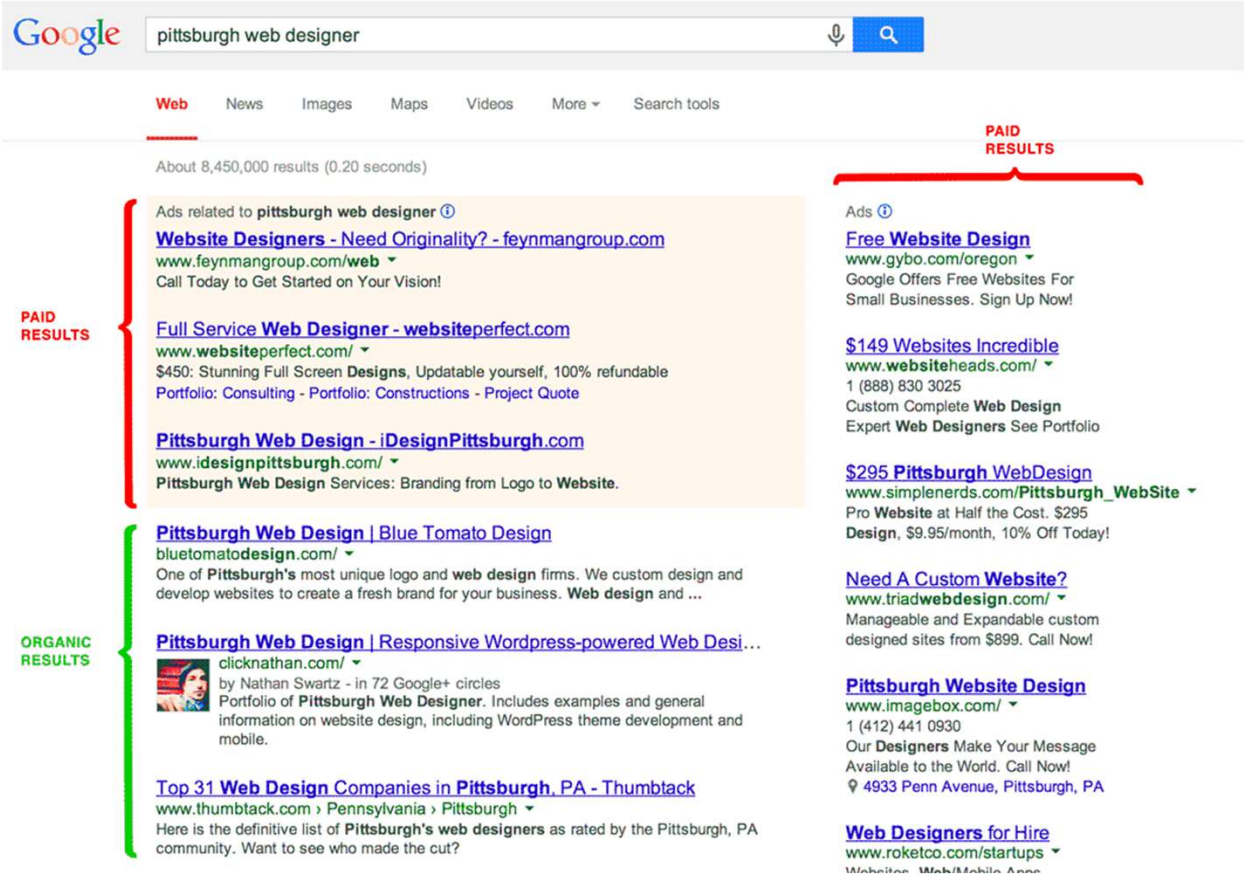
Machine Translation

Trend Analysis

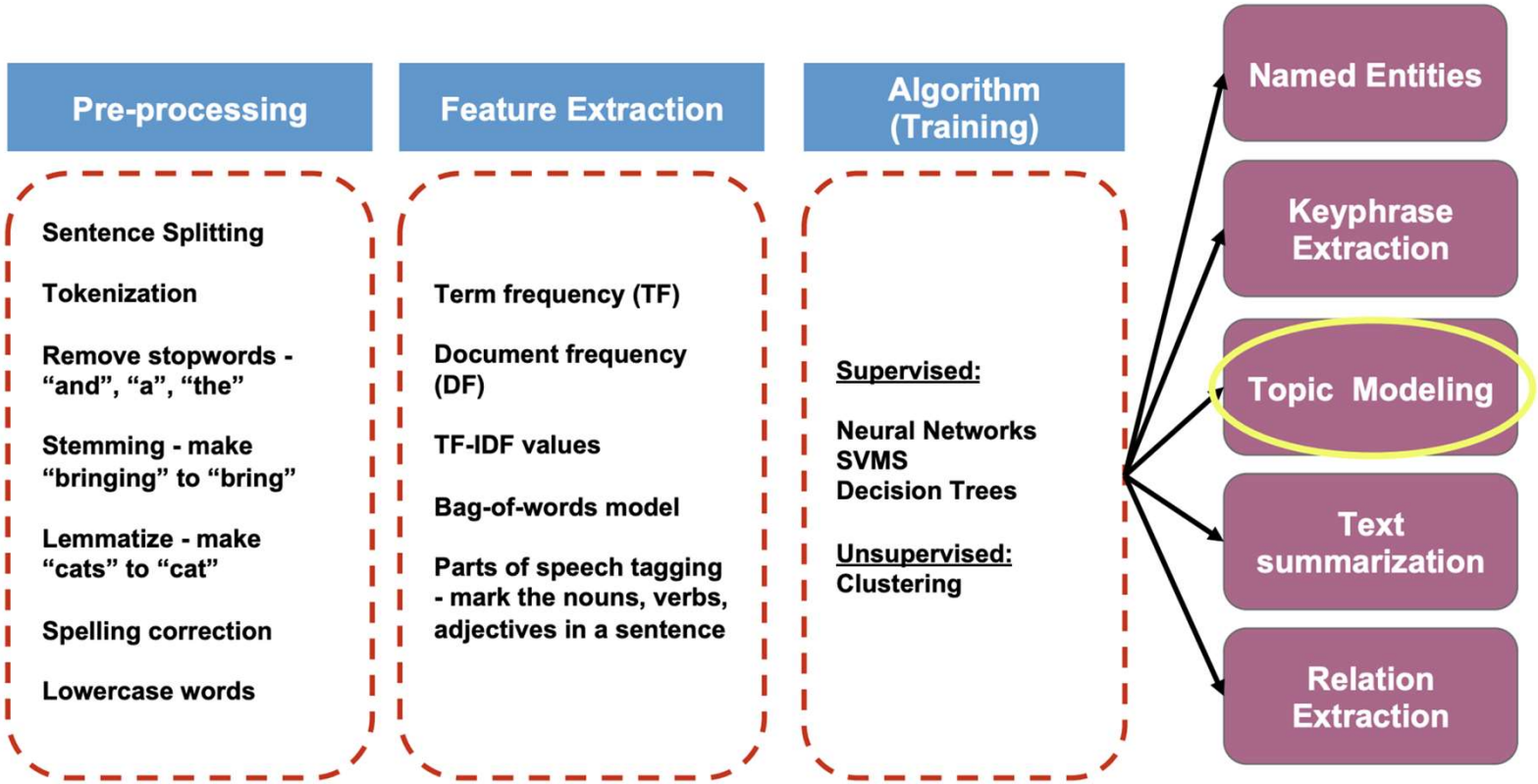
Speech Recognition



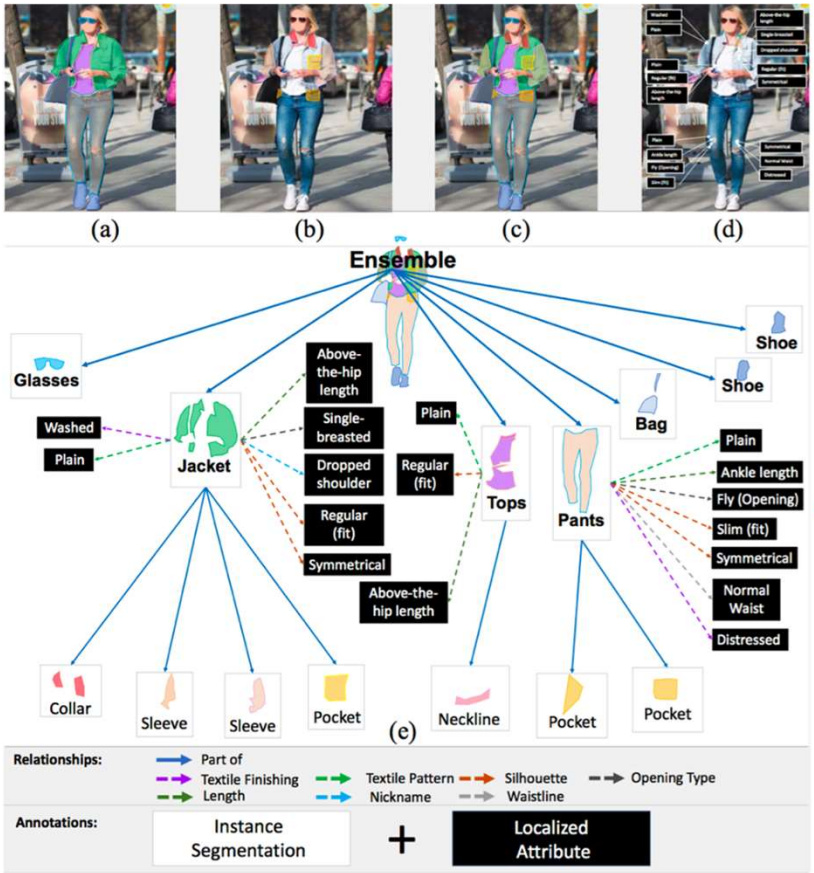
Google Search Result

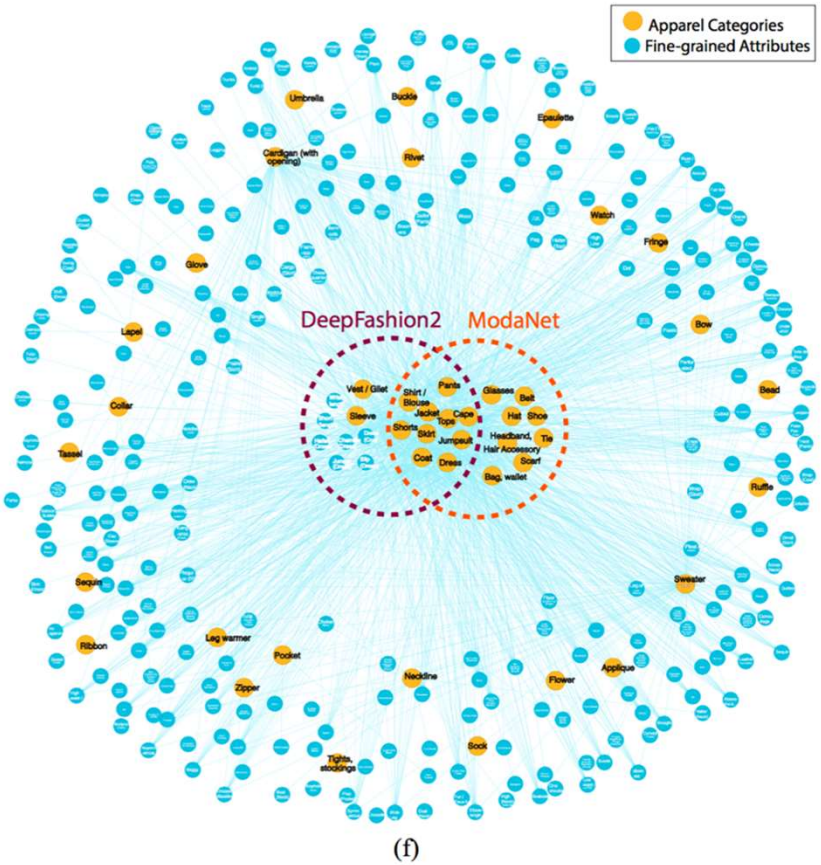


Natural Language Processing

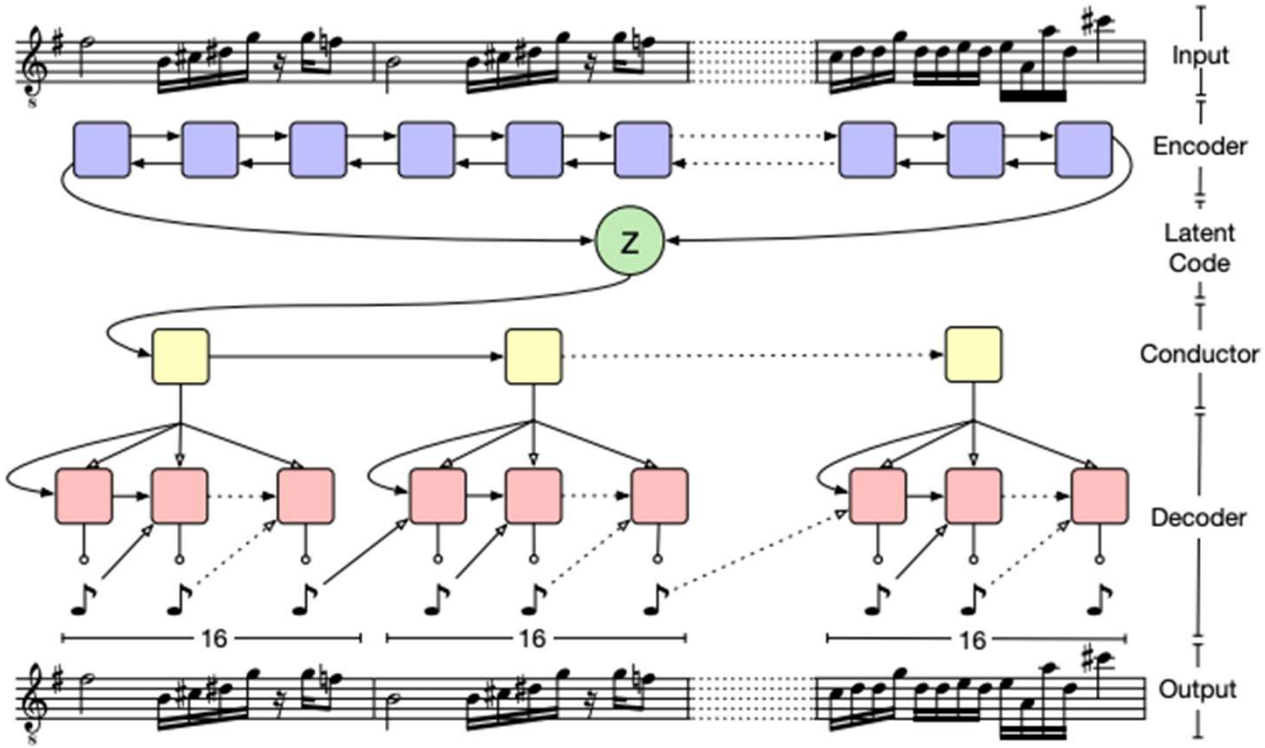


Web Data Mining - Visual (Image, Video)





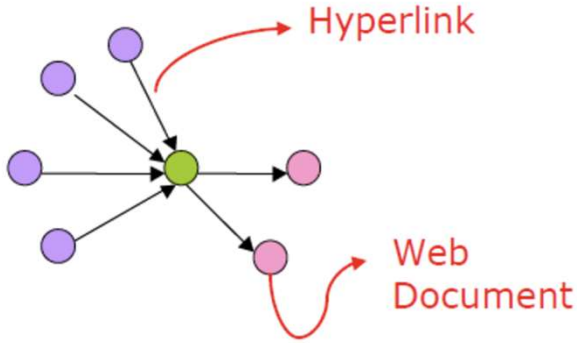
Web Data Mining - Audio



Ref: [Google's MusicLM: Text Generated Music]

Web Structure Mining

- **Goal**: Web structure mining is the process of discovering structure information from the web.
- The structure of typical web graph consists of **Web pages as nodes**, and **hyperlinks as edges** connecting between two related pages.
- This type of mining can be performed either at the **document level (intra-page)** or at the **hyperlink level(inter-page)**
  - PageRank
  - Hubs and Authorities - HITS



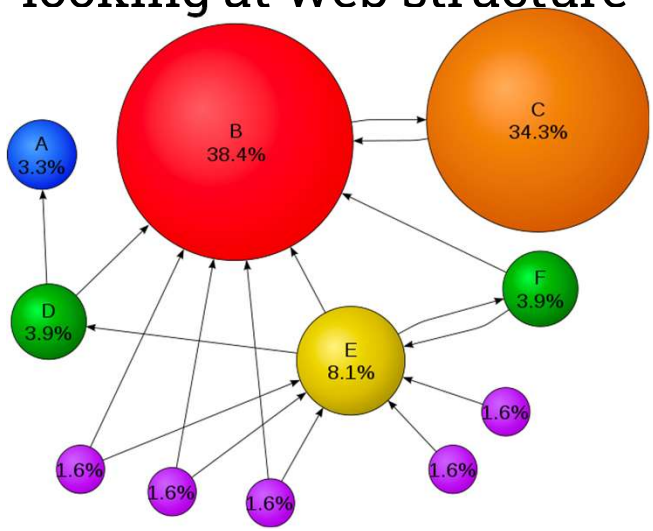
PageRank

- **PageRank (PR)** is an algorithm used by Google Search to rank websites in their search engine results.
  - Used to discover the most important pages on the web.
  - Prioritize pages returned from search by looking at web structure

$$PageRank\ of\ site = \sum \frac{PageRank\ of\ inbound\ link}{Number\ of\ links\ on\ that\ page}$$

OR

$$PR(u) = (1 - d) + d \times \sum \frac{PR(v)}{N(v)}$$



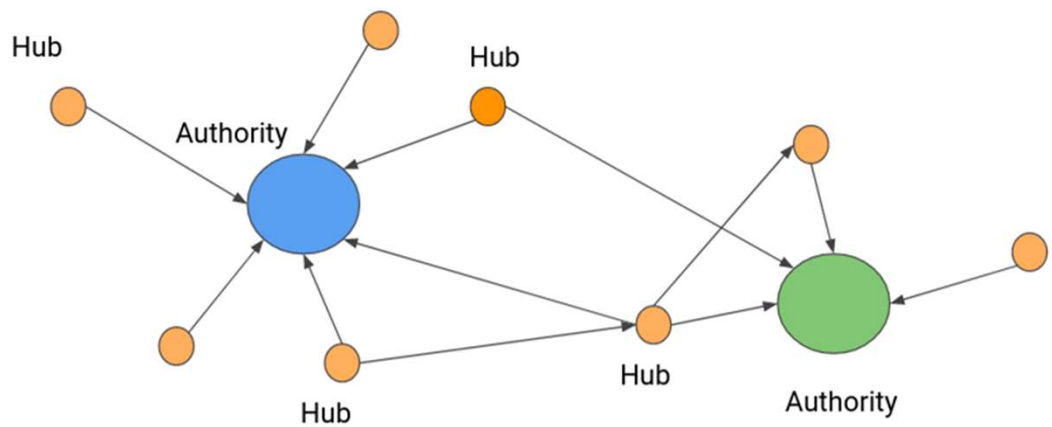
https://en.wikipedia.org/wiki/PageRank

Hubs and Authorities

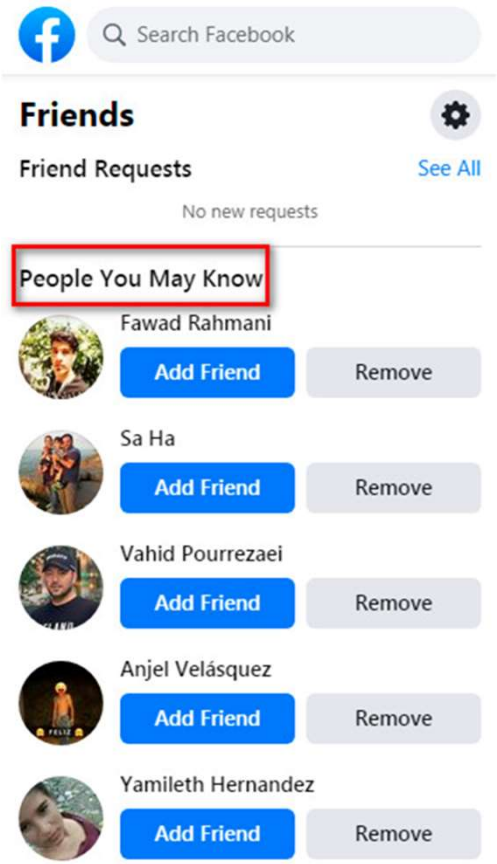
- Authoritative pages
  - Highly important pages.
  - Best source for requested information.
- Hub pages
  - Contains links to highly important pages.

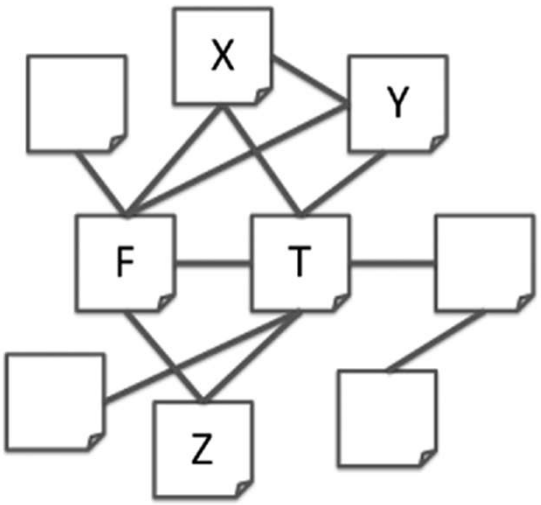
$$Authority(V_i) = \sum_{V_j \in In(V_i)} e_{ji} \cdot Hub(V_j)$$

$$Hub(V_i) = \sum_{V_j \in Out(V_i)} e_{ij} \cdot Authority(V_j)$$



Link Prediction





Citation Network

Web Structure Mining - Applications

- **Quality of Web Page -** Ranking of web pages
- **Interesting Web Structures -** Graph patterns like Co-citation, Social choice, etc.
- **Web Page Classification -** Classifying web pages according to various topics
- **Finding Related Pages -** Given one relevant page, find all related pages
- **Detection of duplicate pages -** Detection of neared-mirror sites to eliminate duplication

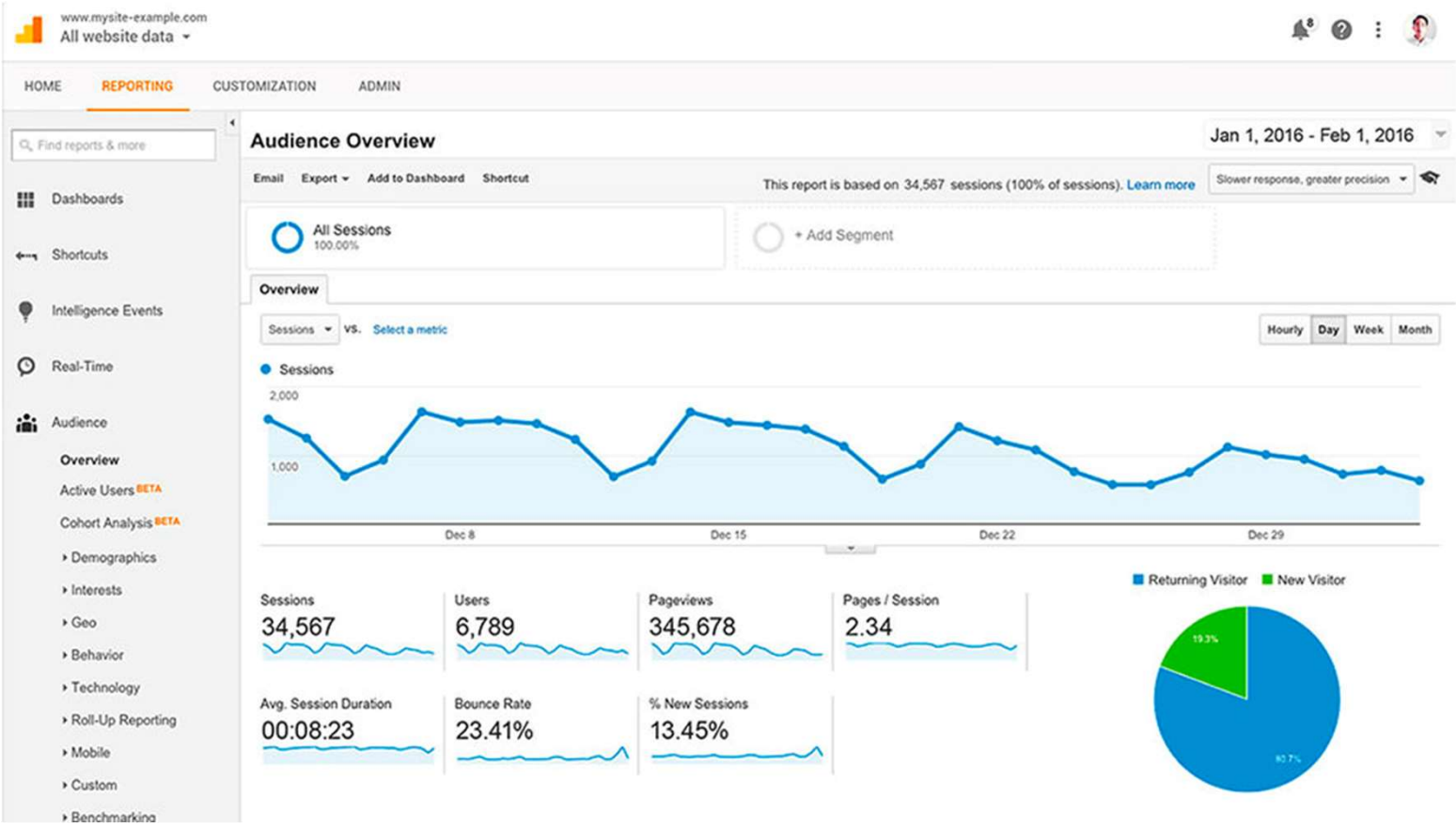
Web Usage Mining

- **Goal**: It is the process of applying data mining techniques to the discovery of **usage patterns** from Web data to **understand** and **better serve the needs** of Web-based applications.
- Data from user interactions with a Web sites:
  - Web logs, click streams and database transaction
- Typical Techniques:
  - Statistical
  - Association Rules
  - Clustering
  - Classification

An Example of Server Log

1	2006-02-01 00:08:43 1.2.3.4 - GET /classes/cs589/papers.html - 200 9221 HTTP/1.1 maya.cs.depaul.edu Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1;+SV1;+.NET+CLR+2.0.50727) http://dataminingresources.blogspot.com/
2	2006-02-01 00:08:46 1.2.3.4 - GET /classes/cs589/papers/cms-tai.pdf - 200 4096 HTTP/1.1 maya.cs.depaul.edu Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1;+SV1;+.NET+CLR+2.0.50727) http://maya.cs.depaul.edu/~classes/cs589/papers.html
3	2006-02-01 08:01:28 2.3.4.5 - GET /classes/ds575/papers/hyperlink.pdf - 200 318814 HTTP/1.1 maya.cs.depaul.edu Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1) http://www.google.com/search?hl=en&lr=&q=hyperlink+analysis+for+the+web+survey
4	2006-02-02 19:34:45 3.4.5.6 - GET /classes/cs480/announce.html - 200 3794 HTTP/1.1 maya.cs.depaul.edu Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1;+SV1) http://maya.cs.depaul.edu/~classes/cs480/
5	2006-02-02 19:34:45 3.4.5.6 - GET /classes/cs480/styles2.css - 200 1636 HTTP/1.1 maya.cs.depaul.edu Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1;+SV1) http://maya.cs.depaul.edu/~classes/cs480/announce.html
6	2006-02-02 19:34:45 3.4.5.6 - GET /classes/cs480/header.gif - 200 6027 HTTP/1.1 maya.cs.depaul.edu Mozilla/4.0+(compatible;+MSIE+6.0;+Windows+NT+5.1;+SV1) http://maya.cs.depaul.edu/~classes/cs480/announce.html

Web Usage Mining - Statistics



Web Usage Mining - Association Rules

ID	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}
…	…

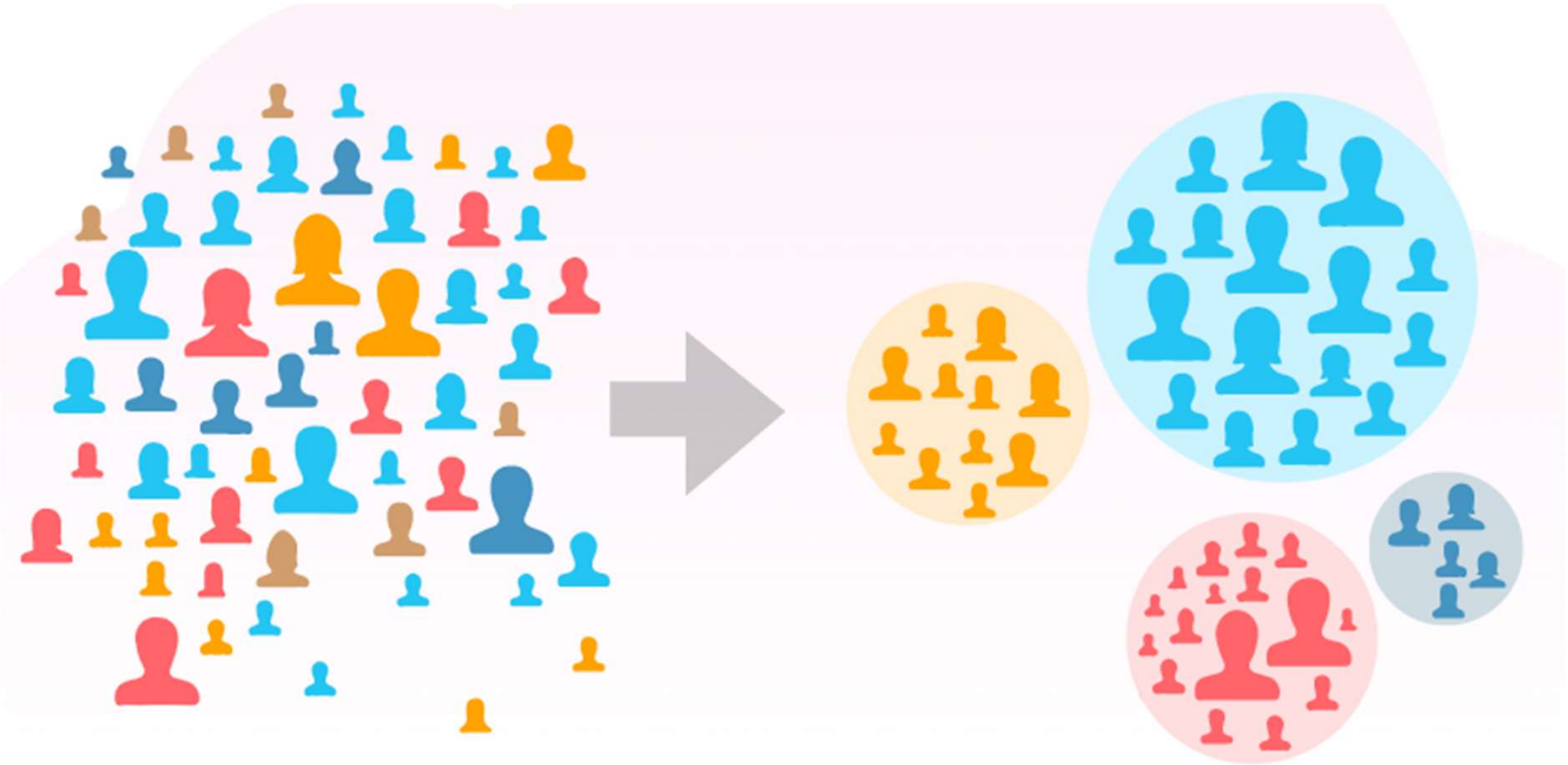
market basket transactions

{Diapers, Beer} Example of a frequent itemset

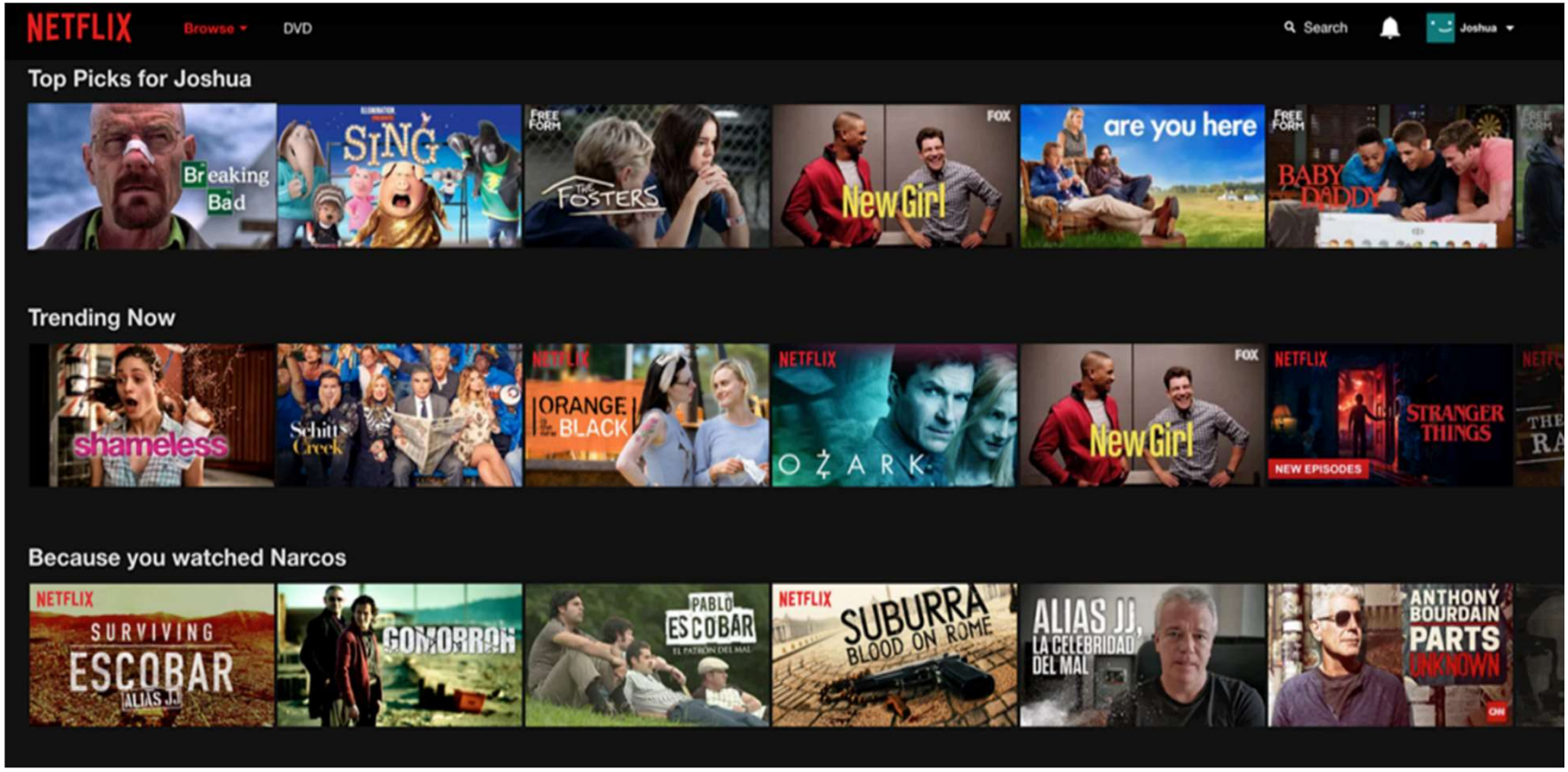
{Diapers} → {Beer} Example of an association rule

Algorithms: Apriori, FP-Growth, …

Web Usage Mining - Clustering/Classification



An Example of Recommendation



Web Usage Mining - Applications

- **Recommender systems -** Suggest customers with relevant items from million ones.
- **User Profiling -** Uncover the information about user profiles
- **Web marketing & targeted advertising -** Reach to right customers
- **Personalized search -** Retrieve better results.

Summary

- Web Mining Overview
- Web Content Mining
- Web Structure Mining
- Web Usage Mining

Thank you

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